



Twitter as a predictive system: A systematic literature review

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ABSTRACT

Millions of people use Twitter daily, posting thousands of messages and interacting with their peers. This research aims to evaluate and classify the predictive potential of the Twitter social platform through the intelligent analysis of user-generated public big data analytics. A systematic literature review (SLR) covering Web of Science, IEEE, Scopus and other databases identified the gaps and opportunities for developing predictive applications of User-Generated Content (UGC) on Twitter since 2006. Our research is a practical contribution to the use of Twitter data as a predictive system. A wide variety of application domains, highlighting social network analysis and public health, have been identified by applying innovative techniques for conducting a massive SLR, leveraging machine learning and graph analysis. The results give rise to new research lines with implications for both scholars and business leaders.

1. Introduction

The information revolution has given rise to new economies structured around flows of data, information and knowledge (Serrat, 2017). Content consumption has radically changed with the development of the Internet. The large volumes of generated data, the rapid dissemination of information, and their easy accessibility have made social platforms a perfect example of interaction between millions of individuals (Gadek et al., 2018): people who engage daily with the content published, making their networks successful information exchange platforms (Wang et al., 2010).

As Streeter and Gillespie (1993) point out, a social network is defined by a finite set of connected social units; therefore, if we wish to state who is and who is not a member, we must define the criteria for membership. Thus connectivity—the links between individuals and the social units or groups that interact with them—is a crucial element in the analysis of social networks (Wasserman & Faust, 1994).

Most of the data generated in social networks are unstructured (Gandomi & Haider, 2015) and therefore difficult to measure, so it is analysed through a range of characteristic features. Twitter offers a large amount of unstructured and voluminous data generated from heterogeneous sources quickly (Udanor et al., 2016). Therefore researchers and organisations are interested in leveraging this information to understand better the dynamic changes and trends in the environment (society, market, etc.) that are identifiable in this data. Businesses hope thereby to minimise the risks and impacts of these changes and to

maximise their benefits and opportunities. As a result of digital transformation, the advent of computational intelligence has enabled advanced analysis of social networks in recent years (Nedjah et al., 2022) due to the big data nature of Twitter, characterised by its variety and volume, and the speed with which is generated (Casado & Younas, 2015). The evolution of artificial intelligence (AI)-based technologies is rapidly opening up a new frontier in the analysis of massively generated data, supported by the recent and continuous development of supercomputing power (Duan et al., 2019). It is also paving the way for new research due to its general applicability to different domains, from economics and politics to healthcare and marketing. Therefore, given the above, an evaluation of Twitter is proposed as a latent predictive system.

SLRs related to the use of Twitter have been carried out since 2012, when Aarts et al. (2012) published the first review on Twitter, to the best of our knowledge. He evaluated the state-of-the-art at that time, identifying the characteristics of the online social platforms as actor (as an individual and as a member of a community), message (and the value of the metrics linked to a tweet) and network (information dissemination, social network analysis and influence). Samaras et al. (2020) focus on how Information and Communication Technologies (ICT) have influenced the evolution of intelligent healthcare systems, like using Twitter as a real-time data source for raising awareness on epidemics and contributing to knowledge transfer in both patients and the medical community. Guraya (2016) analysed both Twitter and Facebook to understand the use of these platforms in education, and Smith and Lambert

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(2014), focused on the use of Twitter in healthcare higher education. Giustini et al. (2018) concluded that Twitter was an effective communication channel in public health and medicine after conducting a systematic review of systematic reviews. In this previous research, the goal was not to assess Twitter as a predictive system. Moreover, the scope of those studies is limited: few publications were reviewed, and they were concentrated on a specific area of knowledge, such as healthcare. The work of Kumar and Jaiswal (2020) reflected the increasing academic interest and a wide variety of domains; they evaluated the opportunities, trends and research gaps derived from the application of sentiment analysis and soft computing techniques. But they excluded the existing research about Twitter and its predictive capacity, which uses different techniques from the supervised machine learning algorithms for sentiment analysis.

The present study, therefore, given the lack of systematic reviews from a global perspective on the predictive capacity of Twitter analytics, aims to identify the research and business applications and opportunities derived from its analysis. The research questions are:

RQ1. How much research has there been about using Twitter as a predictive system?

RQ2. What main topics are covered by empirical studies on using Twitter as a predictive system?

RQ3. What applications of Twitter as a predictive system are proposed in the literature?

RQ4. Which are the most impactful journals and conferences?

RQ5. How has the interest in Twitter as a predictive system evolved?

Concerning RQ1, a general analysis was conducted, identifying the relevant papers and their distribution per year by application domain. To answer RQ2, an automatic analysis for topic identification was performed by applying two different methods: Latent Dirichlet Allocation (LDA) and text-to-network analysis, comparing the results. For RQ3, a sub-topic identification was conducted by applying text-to-network analysis, thus identifying the use case categories and the main techniques used. Following this, in order to answer RQ4, the journals and conferences in which the most relevant research has been published were analysed. Finally, concerning RQ5, an account of the evolution of scientific interest is presented, based on the number of citations, identifying the most impactful publications and the ones with the highest potential to become impactful.

The structure of the document continues as follows. A theoretical framework is presented in Section 2 to underpin the study with a literature review. The developed methodologies are presented in Section 3, and the results for the addressed research questions are explained in Section 4. In Section 5, the main insights and results are discussed. Last, Section 6 offers the main conclusions and outlines the opportunities for future research and the limitation of the study.

2. Theoretical framework

The rise of new technological advances in terms of big data analysis and the development of artificial intelligence techniques (machine learning, natural language processing, etc.) is continuously generating new research, whose potential and increasing academic interest in its application to Twitter analytics were earlier identified by Goonetilleke et al. (2014). However, interest is also rising, in organisations across industries, in the translation of the generated data into valuable insights that enable the development of data-driven strategies facilitating the decision-making processes and increasing the efficiency in their operations (Saggi & Jain, 2018). Every minute, 347,200 tweets are shared on Twitter (Domo, 2022), which reflects how the incremental use of social platforms is generating an enormous amount of public data. The analysis of this data can lead to solutions of unresolved problems in different domains or even beyond, generating new services and opportunities (Dong & Lian, 2021) and helping businesses grasp existing opportunities and unmet needs.

The potential of Twitter relies on the aggregated analysis of this

volume of unstructured data in the form of text. The analysis of unstructured data sheds light on the hidden and latent knowledge contained in this increasing amount of available information. Unstructured data is complex to process, as this requires the structural organisation of data for computational processing and analysis (Tanwar et al., 2015). From a business perspective, this complexly interpretable data needs to be translated into business insights and competitive advantages of various kinds. These include improved decision-making process (Blumberg & Atre, 2003), enhanced understanding of customer requirements, increased productivity (Niu et al., 2021) and, for firms, increased performance from management, technology, and talent capability (Aker et al., 2016). Nowadays, the remarkable evolution of AI (sentiment analysis, semantic analysis, topic modelling, etc.) and cloud computing is facilitating the processing of these big data analytics, consequently generating knowledge and value (Moreno & Redondo, 2016).

The present study focuses on the research and business applications and opportunities derived from the analysis of Twitter data and its predictive potential. Therefore, the following categories were considered for Twitter, inspired by the research of Kursuncu (2019):

- **Sentiment analysis, popularity and anomaly prediction:** the publications under this category focus the research on the development and use of sentiment analysis methods, information diffusion model identification (Naveed et al., 2011), popularity prediction (Hong et al., 2011), fake news detection (Buntain and Golbeck, 2017), hashtag classification and prediction, recommendations, TV and films audience (Rui et al., 2013) and the virality of content. This category has the highest number of publications in scope, highlighting the relevance of the sentiment analysis technique and its predictive potential applied to Twitter analytics.
- **Communities on Social Media:** this group includes the research on identifying communities in Twitter using social network analysis techniques (e.g. graph theory). Given the relevance of the studies in this line of research, it is worth mentioning the work developed by two sets of authors. Pei et al. (2014) predicted the information dissemination media with network parameters, building an effective strategy for detecting viral information. And van den Broek et al. (2019) studied the network structure and the culture around 'Movember', a yearly fundraising campaign to create awareness about cancer in men. This category also includes topic identification and research. In this line of research, Gadek et al. (2017) researched topical cohesion in communities by applying graph algorithms.
- **Healthcare & Public Health:** this application domain contains the use of Twitter as a predictive system in healthcare. This varies from outbreak detection and epidemic and pandemic evolution and control, as in the case of Ebola (Khatua et al., 2019), Zika (McGough et al., 2017), Covid-19 (Bisanzio, 2020; Li and L. x. 2020) or flu (Signorini & Segre, 2011; Paul & Dredze, 2014) to public health management, such as mental diseases detection (McClellan, 2017; Sho Tsugawa, 2015) and the detection of adverse drug reactions (Wu, 2015; Moh, 2017).
- **Sales & Stock Prediction:** in this category, the research highlighted has looked at Twitter as a predictive system in finance, credit score (Metz, 2014), cryptocurrencies (Shen et al. 2019; Rouhani & Abedin, 2019), exchange rates (Maqsood et al., 2020), stock price prediction (Bollen et al., 2011), product and company sales (Kim and Kim, 2018; Chu & Sung, 2015), product claims and charity donations (Korolov et al., 2016).
- **Society:** this category includes studies of the individual (predictions of behaviour, user profiling and the detection of emotions and feelings) (Golbeck et al., 2011) and social problems such as terrorism, sexism, civil unrest or crime (Gerber. 2014; Burnap et al., 2016).
- **Politics:** this domain includes research studies related to politics. One especially important example is Tumasjan et al. (2011), who

used aggregated public opinion to predict the results of elections in Germany. Another is the classification of users' political affiliations.

- **Energy, Environment and Natural Disaster Management:** Twitter is also used to detect rapidly that a natural disaster (like an earthquake, hurricane, flood, tropical storm, etc.) is taking place. This allows the fast mobilisation of the emergency services to minimise the disaster's impact on the population, as in the case of [Neppalli et al. \(2017\)](#), who applied sentiment analysis in the emergency response to a hurricane in the US. Other authors, such as [Kim and Hastak \(2018\)](#), used thousands of tweets to measure how these crises are managed. Recently, [Kirilenko et al. \(2015\)](#) evaluated the use of mass media as a sensor to measure public opinion in the US and its relationship with the increase of the local temperature due to climate change, which reflects the degree of progress toward a more sustainable society ([Bouncken et al., 2022](#)).
- **Location Estimation:** here, several scientific publications have explored Twitter's capacity to predict the location of a user based on the GPS information contained in the tweet and by text analysis or language detection ([Han et al., 2014](#); [Takhteyev et al., 2012](#)). It is important to remember that Twitter removed the precise-location tagging on tweets in 2019 ([Twitter, 2019](#)) due to a change in the users' privacy configuration. [Hu and Wang \(2020\)](#) analysed the impact of this change on the research community.
- **Business:** publications in this category summarise how Twitter is a support tool at the enterprise level, identifying and even predicting new ways of doing business ([He et al., 2013](#)), identifying and collecting customer feedback ([Mostafa, 2013](#)) and measuring the performance of digital marketing campaigns ([Saura, 2021](#)).
- **Transportation:** This domain comprises the research activity and analysis from Twitter analytics about transport. This has been used to monitor and predict traffic, as in the case of [Zhang et al. \(2018\)](#), who were able to detect traffic accidents based on aggregated tweets. It includes studies related to mobility pattern detection, highlighted by [Luo et al. \(2016\)](#), who studied the population of Chicago and classified their mobility based on demographic factors. It also includes studies about routes and the uses of public transport.
- **Bibliometrics:** this covers studies that leverage the use of public data from Twitter to measure and predict the impact and relevance of scientific research and publications ([Peoples et al., 2016](#)). One particularly important aspect is its use in predicting Altmetrics, defined by [Priem et al. \(2011\)](#) as alternative impact metrics in research leveraging different sources on the Internet. Another is the research of [Eysenbach \(2011\)](#), who identified a positive correlation between top-tweeted articles and their corresponding citations.
- **Demographics:** Twitter, like any social platform, is a demographic data source, as the users provide such socioeconomic information as employment, education, income and rates of marriage, births and deaths. [Udayakumar et al. \(2018\)](#) studied tweets related to Psychology publications in order to categorise users by demographic variables. In the case of gender prediction, [Roy et al. \(2021\)](#) used big data technologies to predict the gender of users, given their tweets.
- **Sports:** this includes the use of Twitter to predict sports results such as football ([Brown et al., 2018](#)), basketball or cricket, based on user-generated content. It also includes the analysis of the performance of professional sports players based on their Twitter activity, as [Koster and Aven \(2018\)](#) demonstrated empirically with NBA basketball players.
- **Language:** the studies that look at the use of Twitter as a data source of natural written language in order to develop more precise natural language techniques applied in sarcasm and irony identification ([Ling & Klinger, 2016](#)). Also, authors have used sentiment analysis to predict the emoji evoked by a tweet ([Wu et al., 2018](#)), detect lexical changes in the language ([Eisenstein et al., 2014](#)), extract entities in multi-language sentences ([Sharma et al., 2021b](#)) and identify specific vocabulary uses ([McKay, 2020](#)).

- **Others:** uncategorised publications are set as "Others". Examples include, in Twitter, identifying and predicting cyberattacks ([Hernández-Suárez et al., 2018](#)) the use of Twitter data in analysing and evaluating workers' opinions on the acceptability of robots and artificial intelligence in a shared workplace and the impact of the technology on their jobs ([Sinha et al., 2020](#)).

The goal is to provide a deep, clear vision of the state-of-the-art of Twitter as a predictive system. There is a growing interest in understanding its presence in the academic literature, along with identifying gaps and opportunities for both researchers and business leaders to leverage this data.

3. Methodology

To address the research questions, an SLR was conducted, according to the recommendations of [Kitchenham \(2004\)](#), which emphasise taking into account the importance of identifying the need for the review, identifying the data sources in scope, providing a thorough description of the performed review, presenting the results clearly, and identifying the gaps when communicating the main findings to enable new lines of research. 7,928 documents were collected from 2007, after the launch of Twitter in March 2006, to November 2021.

3.1. Search strategy

Data on scientific papers were gathered from the most relevant sources for scientific publications in computer science: ScienceDirect, Scopus, Web of Science (WoS), Institute of Electrical and Electronics Engineers (IEEE), Google Scholar. Furthermore, the metasearch engine [Dimensions.ai](#) ([Herzog et al., 2020](#)) was combined with the aforementioned databases.

The search terms used in the query of the list of eligible publications in English are contained within the title, keywords or abstract of every publication. They were:

TITLE-ABS-KEY("Twitter" OR tweet*) AND TITLE-ABS-KEY(predict* OR forecast*).

In total, 7,928 publications were collected, covering the period from March 2006, when Twitter was publicly launched, until November 2021. The details are described in [Table 1](#).

3.2. Inclusion and exclusion criteria

Out of the 7,928 documents initially retrieved ([Table 2](#)), 3,463 duplicates and publications without a valid DOI were removed. The remaining 4,465 documents represented 56.32 % of those included in the database. In addition, 1,215 papers—including studies related to the use of Twitter as a system for collecting survey data—were classified as "Not applicable" (N/A), as being out of the scope of the present research. Also excluded were the following: publications related to analysis performed with the Weibo SINA platform (also called "Twitter China") and Facebook, publications not written in English and studies related to other research fields (such as medicine) whose purpose was not to use Twitter as an analytical or predictive system.

As a result of this selection ([Table 2](#)), 4,478 papers were discarded

Table 1
Distribution of publications per source.

Sources	No. of papers
Scopus	1,739
Dimensions.ai	2,271
IEEE	914
Web of Science (WoS)	1,000
Science Direct	2,004
Total	7,928

Table 2

Inclusion criteria for the articles in the scope.

Phase	Criteria	Publications in scope
Database creation	Match search query	7,928
Search results	No duplicates	4,465
	No valid DOI or lack of DOI	
Exclusion of descriptive articles	Not applicable	3,250

from the review, and 3,250 contributions (40.99 % of the database initially created) were selected.

It is also worth mentioning that in order to have a data source that was consistent and harmonised with regard to the relevance of the journals, their corresponding impact factor (RQ4) and the citation evolution analysis (RQ5), the source for the citation data is the Scopus API (Rose & Kitchin, 2019). Some of the articles remaining after exclusions were not indexed in this well-known bibliometric database (Montoya et al., 2018), possibly being conference-related unindexed publications. These (305) articles were also discarded. Therefore, to answer RQ4 and RQ5, 2945 publications remained in the study's scope, representing 90.61 % of the total for these research questions.

3.3. Data collection and analysis

Each of the final 3,250 articles has been tabulated to show:

- Title, abstract and year of publication (addressing RQ1, RQ2 and RQ3).
- Application domain, after manual labelling (addressing RQ1 and RQ2).
- Journal & impact factor, using SCImago as the data source (SCImago, 2020), addressing RQ4.
- The number of cites, authors, affiliations and their countries covering RQ5, updated as of February 23, 2022.

Initially, the publications were manually classified according to their field of research. The classification criterion was the so-called application domain, for which the title and abstract of each article were read and, in case of doubt, also part of the content. The classification was reviewed by all the authors of the present study. However, it is important to mention that only one application domain was assigned per article, so that the predominant topic was chosen over the rest, in case

many application domains are represented in one specific publication, following the categories mentioned above by Kursuncu (2019). In addition, the analysis was reinforced by machine learning and graph analytics, using topic modelling, which, unlike traditional SLRs, allows us to digest a larger number of articles.

4. Results

4.1. How much research has there been about using Twitter as a predictive system?

The distribution of publications by application domain and publication year, in descending order by academic interest according to the aforementioned manual clustering, is shown in Table 3.

In answer to RQ1, Table 3 shows an increasing interest in researching the potential of Twitter as a predictive system, with the highest research activity in 2020 and 2021, according to the number of publications.

The evolution of artificial intelligence (AI) and machine learning, as represented in the case of Sentiment analysis as an NLP technique applied to the massive volume of texts generated in Twitter, and the development of computational techniques, is reflected in the increased interest in the application of algorithms to the identification of communities, relevant topics and individuals in the social networks, as reflected in the top application domains.

Notably, the peak of publications about Healthcare & Public Health occurred in 2020 and 2021, probably due to the coronavirus outbreak and the subsequent pandemic of Covid-19 (Velavan and Meyer, 2020), and the interest in evaluating its impact.

In addition, there was a notable increase in academic research interest in Sales & Stock prediction-related studies, possibly motivated by the burst of attention paid to cryptocurrencies by the general audience in recent years (Beck et al., 2019).

In the case of "Location Estimation" related studies, the removal of the precise-location tagging of tweets is reflected in the decrease of academic interest in 2021; 2020 may still reflect articles in press before that controversial decision by the platform.

4.2. What are the main topics covered by empirical studies on using Twitter as a predictive system?

The Latent Dirichlet Allocation (LDA) algorithm for text classification was applied to a text corpus formed by the set of all abstracts and titles in order to confirm the correct classification of the different application domains mentioned above. LDA is a generative probabilistic

Table 3

Distribution of publications per Year and Application Domain.

Application Domain	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	Total
Sentiment analysis		2	6	27	34	49	63	82	73	109	97	153	87	782
Communities on Social Media		4	8	15	38	51	57	65	76	79	58	94	75	620
Healthcare & Public Health	1		4	4	11	13	27	30	51	59	48	95	133	476
Sales & Stock Prediction	1	1	4	5	9	9	22	37	37	39	50	60	54	328
Society	1		4	9	16	16	16	31	30	36	32	49	31	271
Politics	1	1	3	7	13	19	19	19	33	21	32	38	29	235
Disaster Management		1	2	3	5	10	12	14	11	19	22	31	30	160
Location Estimation		1	2	5	7	11	10	16	16	25	14	22	8	137
Business		1	1	1	4	4	4	7	9	9	13	15	9	77
Transportation						1	3	7	9	6	15	6	10	57
Bibliometrics			1			1	1	2	6	3	6	8	7	35
Others					2	1	2	3	1	3	5	6	2	25
Demographics					1		4	4	1	2	4	1	3	20
Sports						1	3	1	5	3	1	2	1	17
Language				1		1		2	1	1	1	3		10
Total	4	11	35	77	140	187	243	320	359	414	398	583	479	3,250

model for discrete data collections such as text corpora (Blei et al., 2003), used for topic identification. As an unsupervised machine learning technique, it detects text patterns in a corpus and automatically groups word clusters that are closely related (Saura et al., 2021). For this purpose, the text corpus was pre-processed using lemmatisation, which groups words of the same family into a single term (Plisson et al., 2004) and removes stop-words: meaningless words such as articles, pronouns, or prepositions (Hidayatullah et al., 2018). The scope of the stop-words was extended to include specific research-related words like ‘Twitter’, ‘tweet’, ‘abstract’, ‘literature’, ‘review’, ‘social’ and ‘media’, as they occur very frequently without adding, in themselves, any insights. After applying LDA, the *c_v* coherence score is 0.426 while the *u_mass* coherence score is -1.972 for a total of 9 topics, chosen according to the categories previously defined and the maximum *c_v* coherence score per number of topics, as represented in Fig. 1.

According to the results shown in Table 4, the previously labelled application domains (RQ1) are reflected in the keywords identified by the LDA algorithm. The topics are consequently inferred, aligned with the categories defined in RQ1, and sorted by the presence weight of each topic. The results confirm the usefulness of applying a machine-learning algorithm to a large set of papers, allowing us to conduct a massive literature review.

Analogously, a text-to-network analysis, also called keyword co-occurrence methodology (Sayyadi & Raschid, 2013), is carried out, taking the technique developed by Paranyushkin (2019) as the starting point, to compare and complement the results of the LDA algorithm for the topic identification. The keyword co-occurrence methodology starts by creating a text corpus (Toda and Kataoka, 2005) formed by the abstracts and titles of the different papers in the scope of the study. A document consists of the concatenation of the abstract and the title for one given publication. The documents are then pre-processed by removing punctuation marks, harmonising them in lower case letters and removing the stop-words, analogously as in the case of LDA. Then the texts are lemmatised, and the collocations are identified, defined as a group of two (bigram) or more words (*n*-gram) that often appear together in a text and are applicable in the development of NLP (McKeown & Radev, 2000). Thus the text-to-network graph is generated, considering each of the unique lemmas as the nodes and the co-occurrences as the edges; the edge weight decreases from three to one while the distance between lemmas increases from one to three. Before the topic identification, the most influential keywords are extracted by using the betweenness centrality parameter (Brandes, 2001), which measures how the information flows through a node and its intermediary influence, along with the identification of the communities formed by nodes that are densely connected by calculating their modularity class (Blondel et al., 2008). Finally, the topics are inferred, analogously as in LDA, for each community and extracting the keywords whose betweenness centrality is higher. As the last step, to visualise the

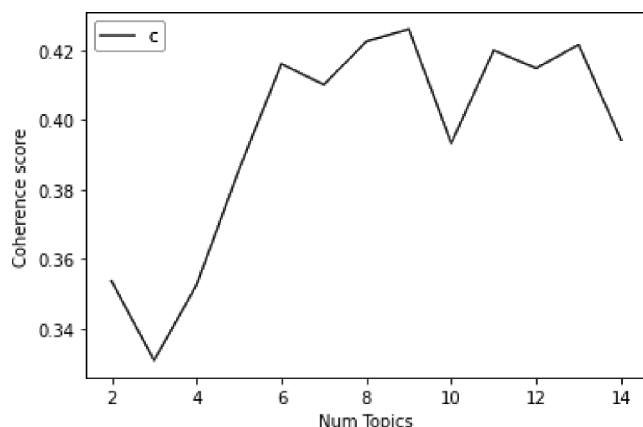


Fig. 1. *C_v* coherence evolution per number of topics in lda algorithm.

Table 4

Latent Dirichlet Allocation (LDA) inferred topics.

R	Inferred Topic	Keywords
1	Twitter as a predictive system	user, network, model, information, predict
2	Location estimation	user, location, time, predict, event
3	Sentiment analysis, popularity and anomaly prediction	sentiment, analysis, feature, language, text
4	Society	model, sentiment, prediction, learning, personality
5	Community on Social Media	use, communication, message, engagement, community
6	Sales & Stock prediction	stock, market, sentiment, predict, price
7	Healthcare & Public Health	Covid, sentiment, public, analysis, health
8	Politics	election, political, predict, result, candidate
9	Bibliometrics	article, impact, citation, Altmetrics, journal

network, the Multilevel Yifan-Hu algorithm is applied (Fig. 2) in Gephi, visualisation and analytical software for graphs and networks (Bastian et al., 2009). Multilevel Yifan-Hu, from the family of force-directed graph layout algorithms, models the network graph as a physical system of particles affected by attraction and repulsion forces between the nodes: the algorithm, therefore, aims to minimise the energy of the system (Hu, 2006). In addition, in order to prevent tag overlapping, Gephi’s “label adjust” algorithm was applied for the final representation of the graph. Finally, the network was filtered by node degree higher than 250 and edge weight higher than 50. Node and label sizes vary according to the node’s betweenness centrality, while the colours represent the different modularity classes.

In Table 5, the topics are inferred based on the top keywords, ordered by betweenness centrality, where the weight column represents the cumulated betweenness centrality of the different lemmas (nodes), expressed as a % of the total:

Analogously, as in the case of LDA (Table 4), in Table 5, most of the topics reflected in most papers (RQ1) were identified by applying the keyword co-occurrence methodology. The utility of the application of this computational techniques relies on its ability to process a large number of research papers, which in turn facilitates an overview of the state-of-the-art and the identification of the most relevant topics and areas of interest in one specific domain. This method could be used in parallel in conducting bibliometric studies and SLRs. In addition, it is potentially helpful in developing studies about text corpus generated on the Twitter platform itself.

4.3. What applications of Twitter as a predictive system are proposed in the literature?

The methodology described above for text-to-network analysis in RQ2 can be applied to the network consisting of the combinations of title and abstract for each paper. By then filtering by application domain or area of knowledge, it is possible to infer the subtopics or applications shown in Table 6 for each domain (Paranyushkin, 2019). Note that the sub-topic list was filtered to show the most relevant topics for each domain.

Going deeper into the defined application domains and research areas (RQ1 and RQ2), the goal of RQ3 is to characterise the state-of-the-art in the predictive capacity of Twitter data by identifying the most relevant applications and use cases. The main subtopics per application domain are thus summarised in Table 6, leveraging the developed keyword co-occurrence methodology in RQ2 for topic identification.

In the case of the two categories “Sentiment analysis, popularity and anomaly prediction” and “Communities on Social Media”, the different predictive techniques used with Twitter UGC were clustered. This highlights the use of social network analysis and graph theory for

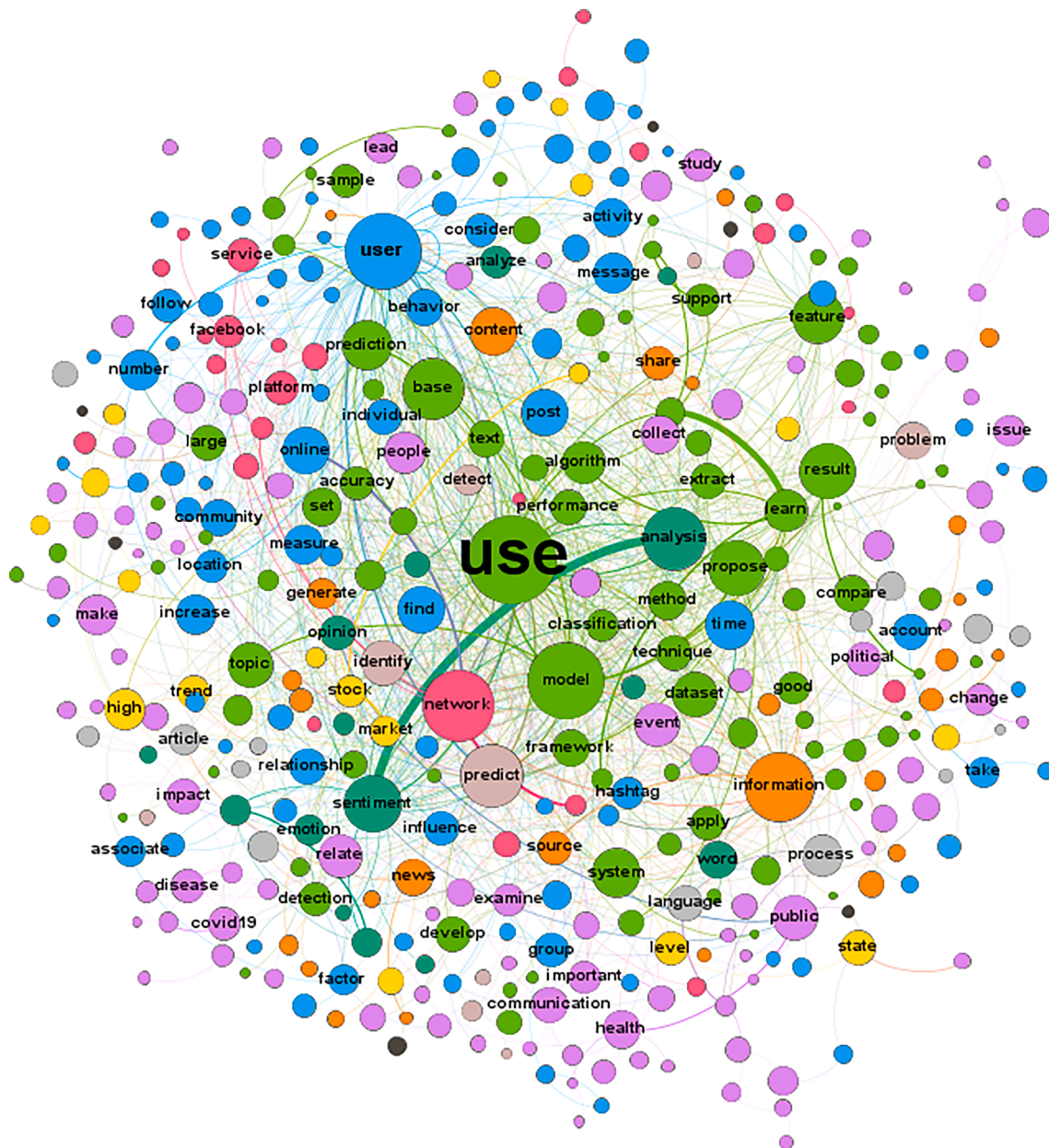


Fig. 2. Abstract & Titles Text-to-Network analysis for topic identification.

community detection and relationship prediction between users. It also demonstrates the use of sentiment analysis for identifying the most popular actors in a given network or predicting the popularity of a given topic, supported by machine learning and natural language processing as main techniques. These techniques are cross-functional: they can be applied in various application domains such as politics, and healthcare, revealing the hidden value in this public UGC.

Research interest in the care of mental health issues is growing. Using big data analytics from Twitter offers opportunities to tackle these problems as a sociological tool, with the social platforms as a reflection of the society. This gives researchers and specialists the advantage of access to large datasets of texts from millions of people, thus affording a better understanding of these diseases and ways of preventing them. In this line of research, [O'dea et al. \(2015\)](#) were able to determine the level of concern evoked by suicide-related tweets and thus help detect and prevent it. Also, in the context of the Covid-19 pandemic and others, this aggregated information can be useful in predicting the outbreak, evolution and impact of disease and epidemics. [Santillana et al. \(2015\)](#) used

the real-time character of tweets to create an ensemble method which consolidated different data sources, including Twitter and predicted the evolution of the flu one week earlier than other techniques.

The exchange of information, *per se*, and the speed at which it is exchanged, have enabled a new digital economy requiring firms to innovate and invest in knowledge in order to ensure their survival (Audretsch et al., 2020). This is reflected in the case of Sales and Stocks prediction, whose primary research areas are stock price prediction, financial forecasting and cryptocurrencies. The research developed by Bollen et al. (2011) is worth mentioning for company stock prediction. Using a Granger causality analysis and a fuzzy neural network, they predicted the closing values of stocks with an accuracy of 86.7 % over a particular period by correlating Twitter feeds with the Dow Jones Industrial Average. In the cryptocurrency market, Mai et al. (2018) studied how social media affect the bitcoin value by developing a sentiment analysis which was later thoroughly investigated and corroborated by Shen et al. (2019) and Ante (2021). These authors demonstrated the impact of Elon Musk's posts on cryptocurrencies, showing how they

Table 5

Text-to-network analysis of articles network's inferred topics.

R	C	Weight (%)	Inferred Topic	Keywords
1		29.87	Twitter as a predictive system	use, prediction, dataset, algorithm, learn
2		16.79	Community on Social Media	user, activity, behaviour, community, relationship
3		11.55	Healthcare & Public Health	public, health, impact, disease, covid19
4		9.01	Twitter as real-time data source Sales & Stocks prediction	source, knowledge, real-time, stream, financial, insight
5		8.22	Sentiment analysis, popularity and anomaly prediction	analysis, sentiment, opinion, investor, polarity
6		6.90	Business Twitter platform	network, platform, product, review, microblog
7		5.71	Sales & Stocks prediction	trend, stock, market, company, return, price
8		5.49	Twitter as a predictive system in Public Health	predict, identify, problem, detect, analyse, depression

significantly affected Dogecoin and threatened the values of freedom of speech and investor protection that blockchain supposedly provided via its decentralisation.

In the case of Politics, the analysis of the aggregated data from Twitter can be applied to predicting an individual's preferred political party, which is potentially valuable for estimating election results. [Colleoni et al. \(2014\)](#) predicted the political orientation of a user in the US (Democrats or Republicans) based on the levels of homophily in the studied networks in Twitter. Political analysis—derived from the effectiveness of political leaders' discourse on social platforms in attracting the attention of society and responses from the international community—is another example of how a wide variety of analyses in different domains can be applied to Twitter. [Georgiadou et al. \(2020\)](#) applied sentiment analysis to estimate the citizens' preferred outcomes in the UK-EU Brexit negotiations, providing a toolbox for decision-makers wishing to reduce the gap between their vision and society's feelings.

4.4. Where are the authors publishing with a more significant impact?

The development of artificial intelligence techniques such as machine learning and natural language processing, along with the rapid growth of big data analytics and the growing use of social network analysis algorithms, has led to increasing research interest in the use of Twitter as a predictive system, due to its potential and its public availability.

As can be seen in [Fig. 3](#), the first publication of a paper using Twitter data as a predictive system appeared about three years after Twitter's creation in 2006. This paper, by [Eysenbach \(2009\)](#), was entitled 'Infodemiology and Infoveillance: Framework for an Emerging Set of Public Health Informatics Methods to Analyse Search, Communication and Publication Behavior on the Internet'. It presented a framework for public health surveillance on the Internet, focusing on Twitter data because of its almost real-time analysis and its utility in detecting and predicting events relevant to health.

Over the years, the interest mentioned above has continuously increased, especially in the case of Q1 and Q2 journals, and journals without impact factor (-). Between them, these three categories represented 90.87 % of the total publications (2,676 out of 2,945 papers in the scope of the analysis). Q1 represented, throughout, the strongest presence with 51.82 % overall (and by far the strongest in recent years), reflecting the relevance of the related research across different domains.

Notably, 784 papers of the 903 (86.8 %) without a journal classification (-) are conference publications. A significant decrease can be detected from 2020, when the trend of previous years was broken, and

Table 6

Inferred topics by application domain.

Application domain	Weight (%)	Inferred Topic	Keywords
Sentiment analysis, popularity and anomaly prediction	17.26	Text analysis	feature, process, text, classification, language
	16.38	Predictive techniques	model, prediction, learn, algorithm, machine
	15.34	Popularity prediction	user, content, online, post, influence, retweet
	10.64	Sentiment analysis	sentiment, analysis, opinion, public, emotion
Community on Social Media	16.61	Community detection techniques	model, algorithm, cluster, graph, classification
	15.22	Prediction on communities	user, predict, node, group, pattern
	12.53	Community detection	community, word, people, analyze, hashtag
	12.25	Social Network analysis	network, influence, structure, dynamic, spread
Healthcare & Public Health	21.21	Mental health	people, individual, patient, population, depression
	14.20	Twitter as predictive system in healthcare	model, predict, learn, method, compare
	11.11	Disease outbreak prediction and analysis	health, pandemic, response, impact, help
	9.10	Pandemic	outbreak, influenza, surveillance, spread, virus
Sales & Stock Prediction	23.23	Stock price prediction	stock, market, predict, price, company
	16.07	Financial forecasting	model, analysis, forecast, performance, compare
	14.60	Cryptocurrencies	factor, cryptocurrencies, event, technology, platform
	13.50	Investment & Sales	sentiment, investor, opinion, public, identify
Society	18.67	Twitter as social data source	use, post, people, word, analyse
	18.50	Study of the individual	predict, personality, behaviour, influence, individual
	7.88	Crime prediction	crime, prediction, increase, news, investigate
Politics	31.15	Election results prediction	election, prediction, politician, electoral, presidential
	21.53	Political communities	political, sentiment, opinion, people, trend
	9.02	Political analysis	candidate, party, vote, lead, speech
Energy, Environment and Natural Disaster Management	10.09	Disaster Management	emergency, communication, response, crisis, fire
	9.30	Disaster prediction	disaster, human, damage, natural, critical
	7.90	Disaster identification Weather forecast	flood, weather, risk, forecast, impact, water
Location Estimation	23.81	User geolocation	location, user, post, content, geographic
	14.25	Prediction on Tourism	use, collect, tourist, mobile, destination
	12.00	Community geo-detection	prediction, method, cluster, geolocation, accuracy

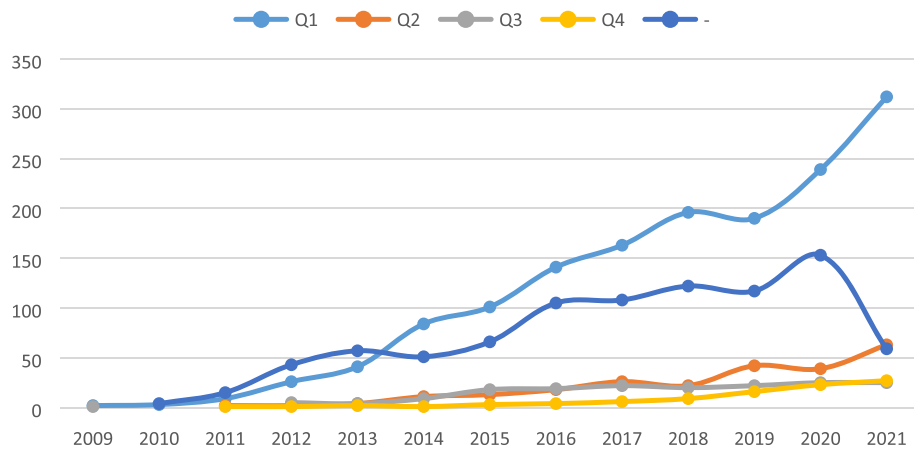


Fig. 3. Evolution of the number of publications in the different quartiles (JCR).

activity reduced from 153 to 59 (-38.56 %) total publications in 2021. This reduction might be explained by the Covid-19 pandemic, which precipitated the cancellation of many conferences and a shift to online events (Moss et al., 2021).

Concerning the number of publications, the most relevant journals are summarised in Table 7, which shows that Expert Systems with Applications (Q1), with 83 articles, and Computers in Human Behavior (Q1), with 79 contributions, are the most relevant journals in the use of Twitter data as a predictive system.

The most representative conference series are Procedia Computer Science, with 110 publications, ACM International Conference Proceeding Series, with 49 articles, and the International Conference on Information and Knowledge Management, with 25 contributions (Table 8).

4.5. How has the interest in Twitter as a predictive system evolved?

In order to identify the publications with the highest research interest, as previously done by scholars such as Ribeiro-Navarrete et al. (2021) and following the contributions of Webster and Watson (2002), Table 9 shows the top 20 most cited articles, according to the WoS database.

Table 9 shows that the most relevant paper is Bollen et al. (2011), previously presented in RQ3. Second is Kalchbrenner et al. (2014), who developed a convolutional neural network for the semantic modelling of tweets with an outperforming efficiency in sentiment prediction. Third is Asur and Huberman (2010), who demonstrated how media content can be used to predict real-world outcomes by developing an experiment to predict movie sales from a model based on the speed at which tweets were created.

In addition, an analysis was conducted of the number of citations, and the impact, of the most relevant journals. The top 5 journals where the most influencing authors are publishing about Twitter as a predictive system are: Expert Systems with Applications (Q1), Computers in Human Behavior (Q1), Information Processing and Management (Q1),

Table 7
Most relevant journals.

Journal	Times Cited	Number of Publications	Impact factor
Expert Systems with Applications	2,864	83	6.954
Computers in Human Behavior	4,443	79	6.829
Information Processing and Management	1,270	56	6.222
Knowledge-Based Systems	1,095	40	8.038
PLoS ONE	2,033	40	3.240

Table 8
Most relevant conferences series.

Conference	Times Cited	Number of Publications
Procedia Computer Science	1,248	110
ACM International Conference Proceeding Series	384	49
International Conference on Information and Knowledge Management	958	25
Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining	779	13

PLoS ONE (Q1) and International Journal of Information Management (Q1). The most representative conferences are Procedia Computer Science, International Conference on Information and Knowledge Management and the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining.

In order to minimise the bias against recently published articles, a rescaling of the citations by publication date, inspired by Boyack et al. (2016), is proposed in Equation (1), which obtains a rescaled yearly cites measure (C_n) that takes into account each article's date of publication:

$$C_n = \frac{365 \cdot \text{citations}}{(\text{Today} - \text{PubDay})} \quad (1)$$

Thanks to the rescaling, recent articles with the potential subsequently to become academically impactful are thus identified in Table 10, which uses the proposed calculation ratio from Equation (1) with the most cited article from Table 9 as a reference in the first row.

Notably, the application domain for all the top five articles in Table 10 is 'Sales & Stock prediction'. This confirms the increased research interest in this area and might also be due to the burst of public interest in cryptocurrencies (Sharma et al., 2021a) and the evolution of machine learning techniques, which are generating new business analyses in economics (Khan et al., 2020). Onan (2021) developed a new deep learning architecture that outperforms traditional deep learning techniques used for sentiment analysis when applied to the classification of product reviews. Similarly, Liu et al. (2021) used NLP to predict customer engagement with a luxury brand and its applicability in developing a social media marketing strategy. Oliveira et al. (2017) focused on forecasting the return on investment, volatility and trading volume variables of company stocks. They combined Twitter data with external data sources and demonstrated the usefulness of this public data for financial expert systems, highlighting its speed, reduced cost and availability. Analogously, Shen et al. (2019) used machine learning algorithms to demonstrate empirically how aggregated Twitter data is useful in predicting the ROI, volatility and trading volume of cryptocurrencies, particularly Bitcoin. Finally, Bartov et al. (2018) developed

Table 9

Top-20 Most cited (WoS) articles.

ID	References	Application domain	Topic	Aim	Times cited (WoS)
S1	(Bollen et al., 2011)	Sales & Stock Prediction	Stock price prediction	Mood analysis in Twitter to predict the stock market with neural networks	2,615
S2	(Kalchbrenner et al., 2014)	Language	Social Network analysis	Language understanding for semantic modelling of sentences	1,393
S3	(Asur & Huberman, 2010)	Sentiment Analysis, popularity & anomaly prediction	Social Network Analysis	Twitter as a predictive system in the real-world	1,101
S4	(Suh et al., 2010)	Sentiment Analysis, popularity & anomaly prediction	Popularity prediction	Aggregated Twitter data to predict retweetability with content features	818
S5	(Signorini et al., 2011)	Healthcare & Public Health	Disease outbreak prediction and analysis	Twitter as a measure of public interest about health	768
S6	(Eysenbach, 2011)	Bibliometrics	Citation prediction	Usefulness to predict the citations of an article based on social media activity	658
S7	(Eysenbach, 2009)	Healthcare & Public Health	Public Health	Twitter activity to predict public health infodemiology	610
S8	(Hughes et al., 2012)	Community on Social Media	Community detection	Relationship between personality and the use of social networking sites	572
S9	(Colleoni et al., 2014)	Politics	Political communities	Predict Political orientation through network analysis	524
S10	(He et al., 2013)	Business	Market analysis	Social media competitive analysis to show the value of text mining	519
S11	(Cheng et al., 2014)	Community on Social Media	Social Network Analysis	Predict content cascades in social networking websites	482
S12	(Smith et al., 2012)	Business	Market analysis	General framework for comparing brand-related user content	479
S13	(Hong et al., 2011)	Sentiment Analysis, popularity & anomaly prediction	Popularity prediction	Twitter stats to classify and predict the popularity of tweets	421
S14	(Schultz et al., 2011)	Community on Social Media	Social Network Analysis	Twitter as medium for crisis communication	419
S15	(Golbeck et al., 2011)	Society	Study of the individual	Personality prediction based on Twitter stats and text analysis	379
S16	(Gerber, 2014)	Society	Social unrest and crime	Crime prediction system based on public data from Twitter	360
S17	(Takteyev et al., 2012)	Location Estimation	Social Network Analysis	Geographic analysis of Twitter network to predict ties between people and places	348
S18	(Achrekar et al., 2011)	Healthcare & Public Health	Disease outbreak prediction and analysis	Seasonal flu epidemics evolution based on public data	326
S19	(Saif et al., 2012)	Sentiment Analysis, popularity & anomaly prediction	Sentiment analysis	Sentiment analysis applied to Twitter data to monitor public opinion	279
S20	(Quercia et al., 2011)	Society	Study of the individual	Personality prediction based on popularity	317

Table 10

Top-5 articles with the highest impact potential.

ID	References	Application domain	Topic	Aim	Times cited (WoS)	C _n
	(Bollen et al., 2011)	Sales & Stock Prediction	Stock price prediction	Mood analysis in Twitter to predict the stock market with neural networks	2,615	237.7
S1	(Onan, 2021)	Sales & Stock Prediction	Investment & Sales	Deep learning techniques applied to product reviews and sentiment analysis	45	207.9
S2	(Liu et al., 2021)	Sales & Stock Prediction	Investment & Sales	Customer engagement analysis with NLP applied to UGC	60	60.3
S3	(Oliveira et al., 2017)	Sales & Stock Prediction	Stock price prediction	Twitter usefulness to forecast stock market variables	152	31.5
S4	(Shen et al., 2019)	Sales & Stock Prediction	Cryptocurrency	Twitter activity as a reflect of trading volume of Bitcoin	94	29.8
S5	(Bartov et al., 2018)	Sales & Stock Prediction	Stock price prediction	Aggregate opinion to assess to future value of a company stock	100	26.1

an extended analysis across different industries and found that a deep statistical analysis of the aggregated opinion from thousands of tweets could predict a firm's quarterly earnings and returns.

5. Discussion

Since its creation, the number of users on Twitter has increased every year and is expected to keep growing (eMarketer, 2020). Accordingly, this systematic review reveals that research interest in using Twitter analytics as a predictive system is also rising. Many articles have been published since Twitter's creation in 2006, especially during 2020, when it peaked. Unlike the process followed in the vast majority of SLRs, we present and compare two innovative methodologies: LDA and text-to-network analysis. In addition to analysing the predictive ability of

Twitter data, these powerful tools allowed us to carry out a massive SLR, analysing the state-of-the-art of a specific area of knowledge by harnessing computational techniques.

As reflected in the revised literature, various techniques have been developed and tested for using Twitter data for predictive purposes. The development of machine learning techniques shows its importance in predicting events with large amounts of public data, highlighting the applicability of Natural Language Processing (NLP) techniques such as sentiment analysis, classification algorithms, clustering and other techniques. This results in the generation of insights across several domains, such as politics, healthcare, sales and stocks and society. In addition, the application of social network and graph analysis to Twitter as a social networking website facilitates understanding of the relationships between the different users and the information generated on the platform.

Combining these computational techniques can, thus, be effectively applied to multiple areas of knowledge leveraging Twitter analytics. For example, the burst of publications in 2020 on Healthcare & Public Health (potentially motivated by the outbreak of the Covid-19 pandemic) is worth mentioning: the number of total publications doubled over the previous year. This area of knowledge is key for academic research since several of the most cited articles fit into this category. Most of the publications in this area are dedicated to detecting, analysing and predicting the outbreak of diseases and their spread, which is essential for epidemic and pandemic management in the public health domain. There is also an open line of research derived from the social dimension of Twitter as a social platform for other related public health topics such as mental health (e.g., suicide, depression and stress management), food habits and addictions. In the area of Sales & Stocks prediction and Business, Twitter has become a valuable tool for both users and companies. Assuming that Twitter users are potential customers and/or investors, there is interest in researching how the aggregated data can be used to predict the sales and stock prices of enterprises in different sectors. A particular example of this is the evolution of the cryptocurrencies market. Analogously, the application of predictive techniques in the area of politics is outlined, specifically its use in forecasting election results and segmenting the population according to their beliefs.

We have seen that the first publications on Twitter were in 2009, and most articles were published as conference proceedings. Regarding the evolution of publications according to their impact factor, it is remarkable that more than half were published in Q1 journals and conferences. Finally, the studies are primarily analytical, based on UGC from Twitter as a public data source, allowing empirical testing of the defined hypothesis.

6. Conclusion

The rapid evolution of AI and machine learning algorithms in recent years has enabled complex analyses that were not previously feasible. In addition, most of the available information in different types of business is unstructured, and there is increasing interest among researchers and business leaders in transforming this data generated in different sources (e.g. metadata, web pages or social media) into connected, structured data that can result in the generation of insights and value.

This review has confirmed the interest in, and the potential of, using Twitter as a predictive system and demonstrated the continuously increasing research activity and the wide variety of techniques and areas of knowledge where its analysis is applied:

- The use of Twitter as a predictive system is extensive. The techniques applied to UGC can be used in many areas of knowledge, from Healthcare & Public Health, Politics and Society to Business or Sales & Stocks, as shown by the high number of publications in these application domains. The ability to predict using these analytics is based on the extended availability of historical and recent information from various global sources.
- The hidden value in the aggregated user-generated content on Twitter is yet to be fully explored. Despite the high number of existing publications about Twitter, there were as yet few studies presenting an overview of the predictive capacity of unstructured and aggregated data in the form of texts (tweets). This study fills this gap and identifies opportunities in different domains, as detailed below in future lines of research.
- Leveraging the application of artificial intelligence and machine learning algorithms, like LDA and NLP, and network analysis techniques, such as text-to-network and graph analysis, allows us to conduct SLRs with many more relevant publications than was previously possible, affording us a broader and deeper view. In the present study, it is worth highlighting that LDA and text-to-network analysis present similar results

- Most of the publications (51.82 %) in the review were published in Q1 journals, showing the academic interest in Twitter as a predictive system, the maturity of the techniques, the topic itself, and its increasing trend over the years.
- Moreover, this review proposes an innovative metric (time normalisation) for minimising the bias on impact factor in traditional bibliometrics that arises from the publication date.

6.1. Implications for practitioners

Leveraging the use of Twitter analytics and its predictive potential is interesting for policymakers, as this data has latent value in a range of areas, such as predicting the results of elections in the field of politics, understanding societal perceptions of specific issues and helping the authorities deal with public health risks and alerts. In addition, business leaders should develop data-driven customer-oriented strategies that ensure value generation. The latent value of this information, publicly available in Twitter, can be harnessed by firms to facilitate decision-making processes and contribute to the improvement of their performance.

In practical business terms, attention should be paid to this data from the point of view of customer satisfaction, marketing success, sales and stock price prediction and the identification of existing trends in markets and society.

6.2. Future research

Future research might profitably perform specific analysis by application domain, highlighting 'Sentiment analysis, popularity and anomaly prediction' and 'Communities on Social Media' in order to develop new or improved analysis and prediction techniques based on public data from Twitter. Going deeper into every application domain, case studies could be undertaken in the 'Healthcare & Public Health' domain: examples could include the Covid-19 pandemic and the social awareness about public health, the detection of adverse drug reactions and patients' perspective in Twitter. Furthermore, as reflected in the myriad of applications and techniques for analysing social media content, the techniques developed in the "Sales and Stocks prediction" and "Business" applications domains could be applied to a cross-functional analysis of the healthcare business market. Many opportunities also remain in the 'Sales and Stocks prediction' domain to apply predictive techniques based on aggregated public data from Twitter so as to predict prices of cryptocurrencies and company stocks more accurately. In the case of politics, the interest is growing likewise. While most of the studies have a local focus, there is the possibility of developing comparisons of the effectiveness of different algorithms, across countries and regions, evaluating why differences occur between areas and the level (local, regional, national) of elections. Such studies might contrast actual results with predicted ones and explore the bias trends when the predictions are inaccurate.

In addition, the network and graph analysis offered here triggers the potential to develop studies about the bibliometric collaboration of authors and affiliations for every application domain in the use of Twitter as a predictive system, with geotagged information from the Scopus API, used here as the journal and citation source database.

To conclude, an analysis would be useful of the ethical concerns for individuals sharing information publicly in social networks and the uses to which that information (e.g., classifying people's political beliefs based on Twitter data) is put.

6.3. Limitations

Overall, this review provides a deep view of the state of the art of Twitter as a predictive system, but it is not without limitations. The review was based on a query built with specific keywords, which limits

the scope of the present research. Different combinations of keywords may lead to slightly different analyses and nuances. Furthermore, in following the best practices defined for SLRs in Kitchenham (2004), the present authors excluded relevant studies where the journals or conferences were either not indexed or out of the scope.

The evaluated data sources aim to be systematic and exhaustive, but new research articles may have been published after the period chosen for the systematic review.

Another limitation arises from the constraints on the studies using public Twitter data. The essential Twitter API enables only research for the previous week and a limited amount of tweets to be extracted. These problems have been dispelled by the new Twitter API for Academic research, released in January 2021, which overcomes these limitations and biases by making available the full historic archive of tweets.

CRediT authorship contribution statement

Enrique Cano-Marin: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Marçal Mora-Cantalops:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Salvador Sánchez-Alonso:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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